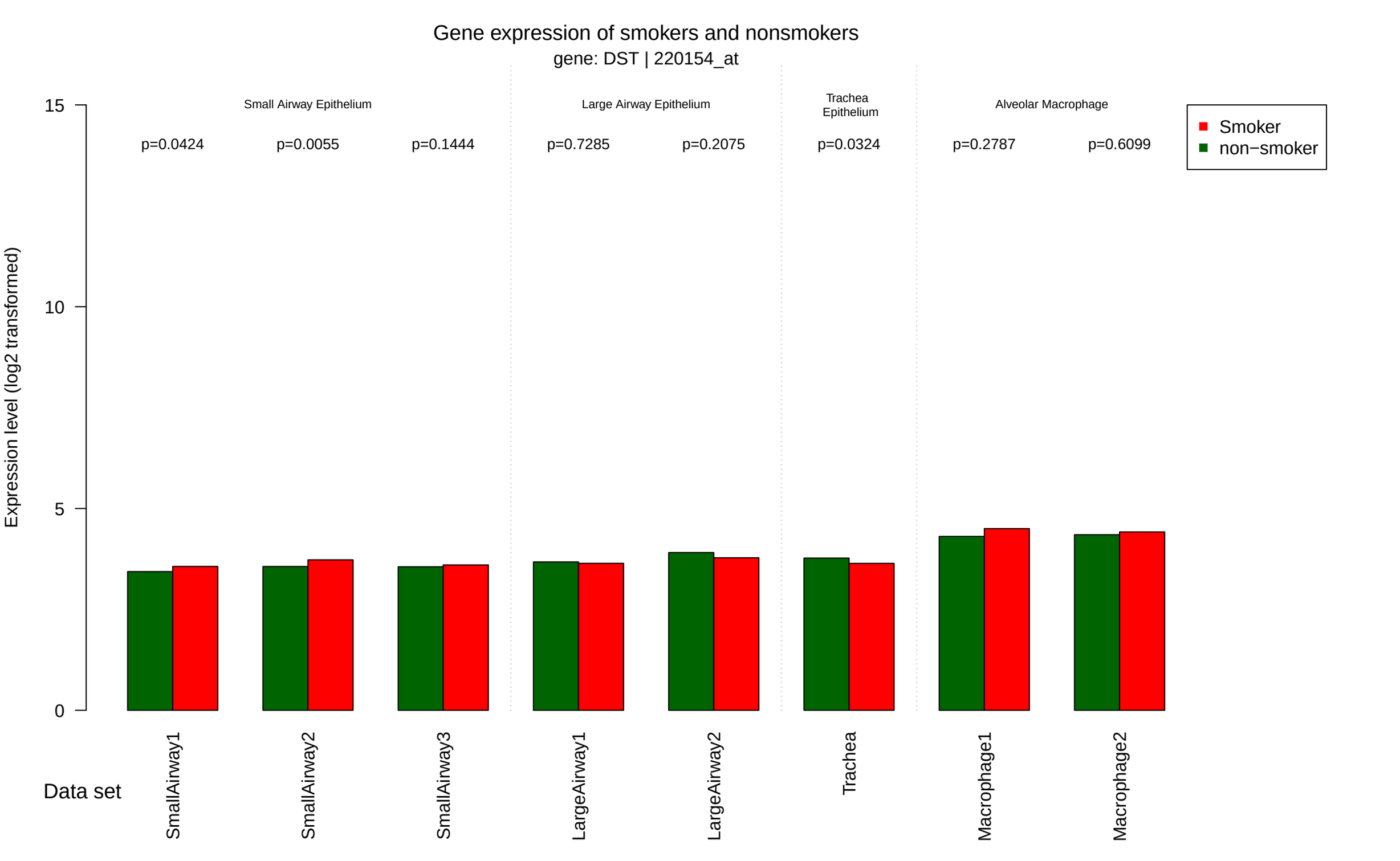
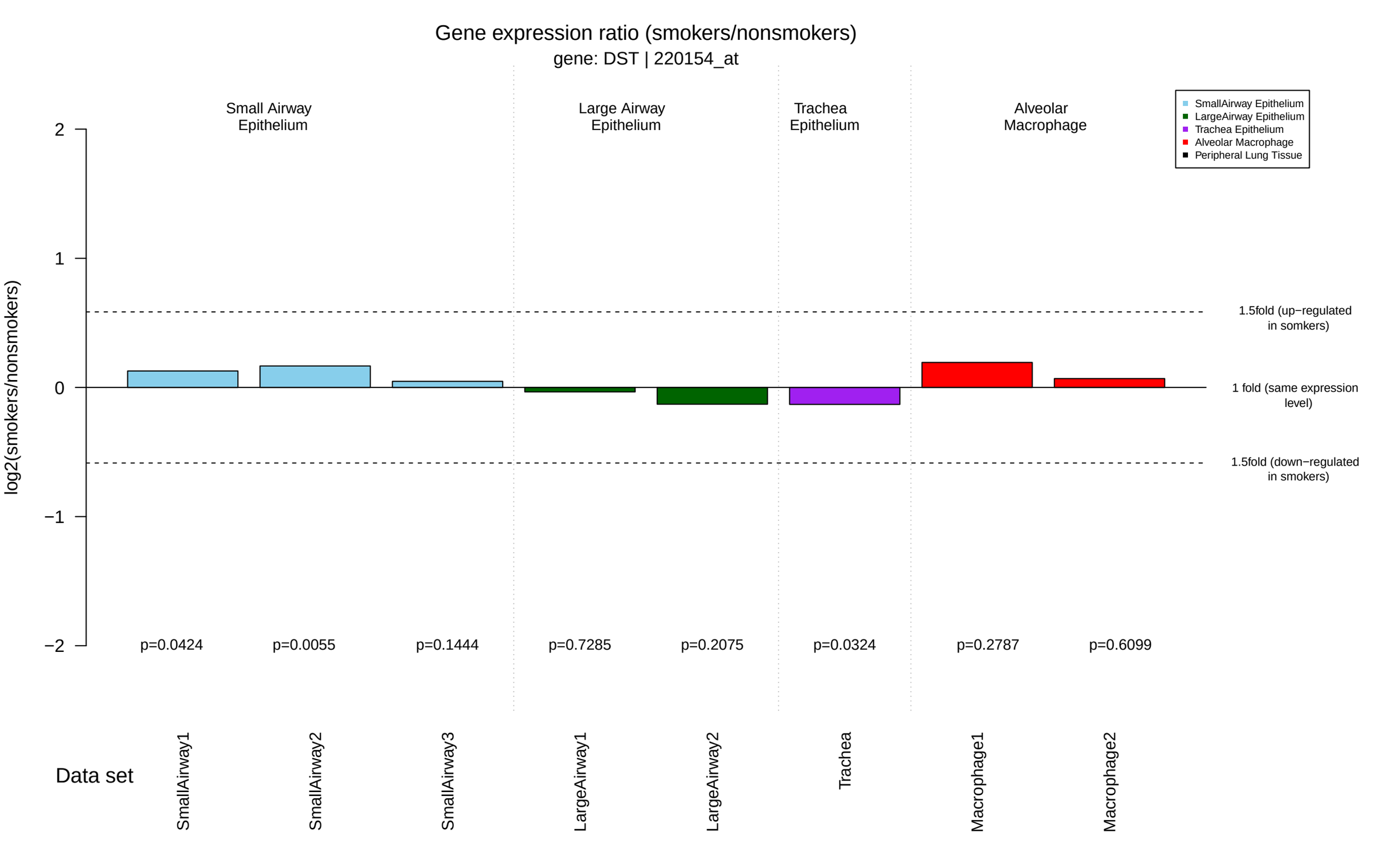
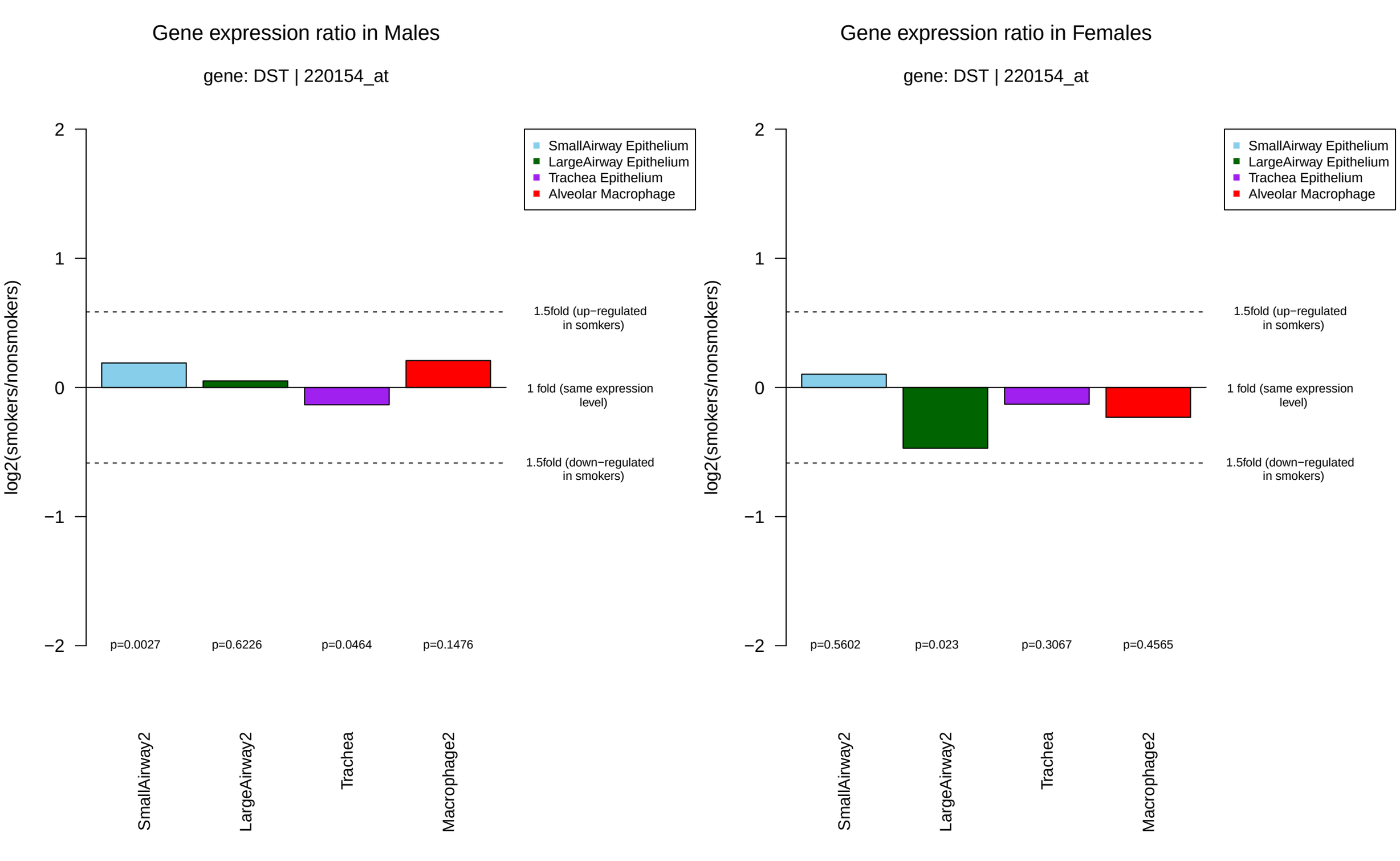
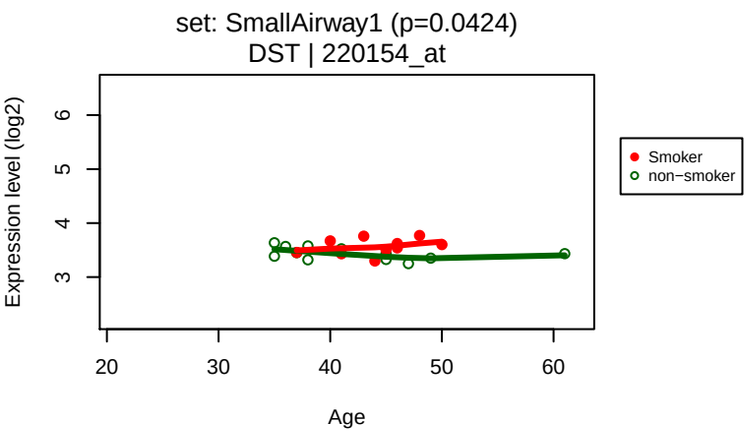
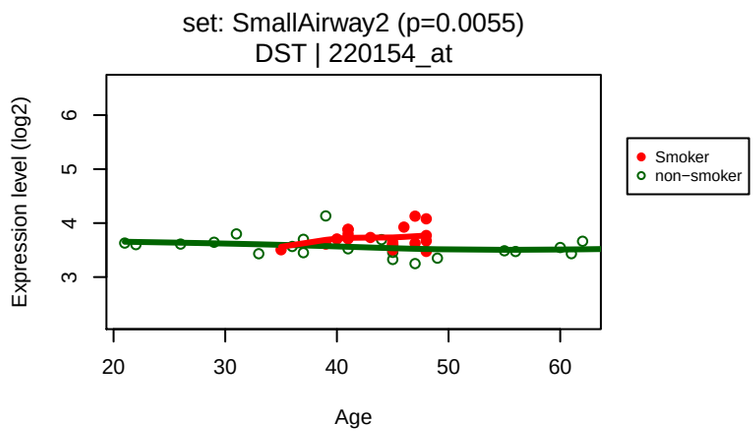
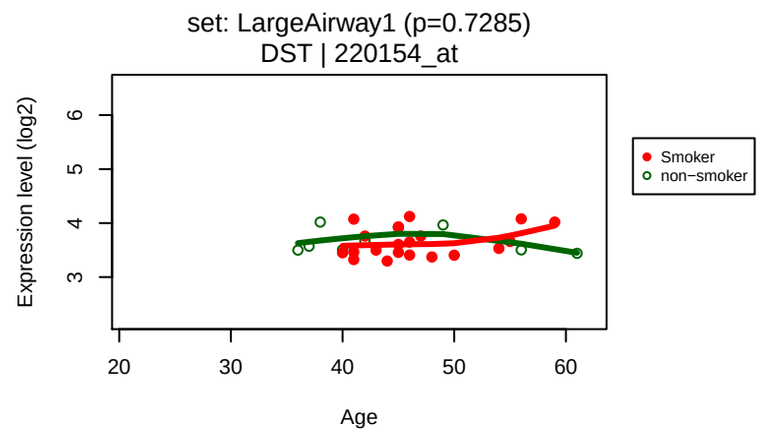

Gene expression of smokers and nonsmokers
gene: DST | 220154_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0424
p=0.0055
p=0.1444
p=0.7285
p=0.2075
p=0.0324
p=0.2787
p=0.6099
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

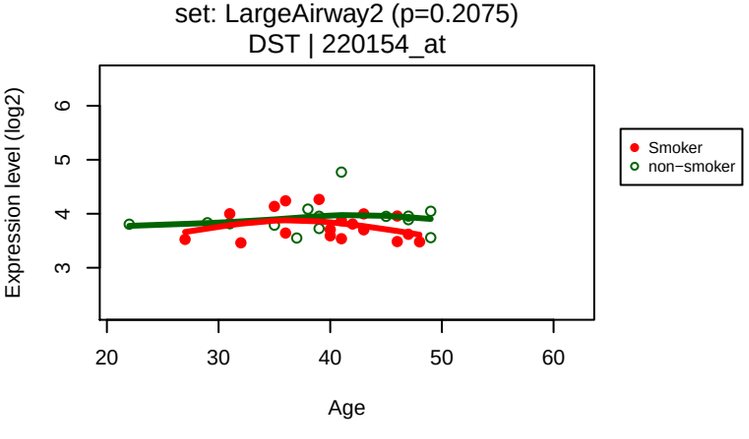

Gene expression ratio (smokers/nonsmokers)
gene: DST | 220154_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.0055
p=0.1444
p=0.7285
p=0.2075
p=0.0324
p=0.2787
p=0.6099
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

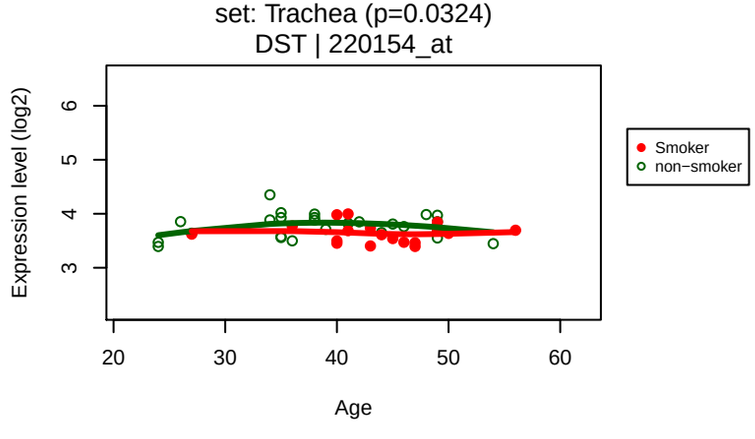

Gene expression ratio in Males
gene: DST | 220154_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0027
p=0.6226
p=0.0464
p=0.1476
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DST | 220154_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5602
p=0.023
p=0.3067
p=0.4565
SmallAirway2
LargeAirway2
Trachea
Macrophage2

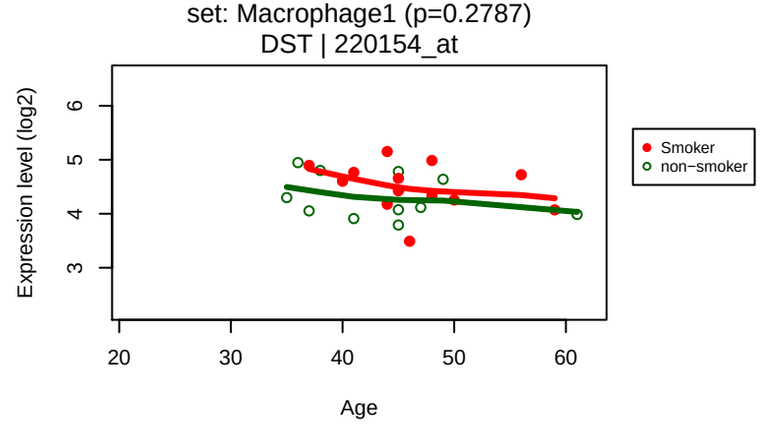

set: SmallAirway1 (p=0.0424)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker

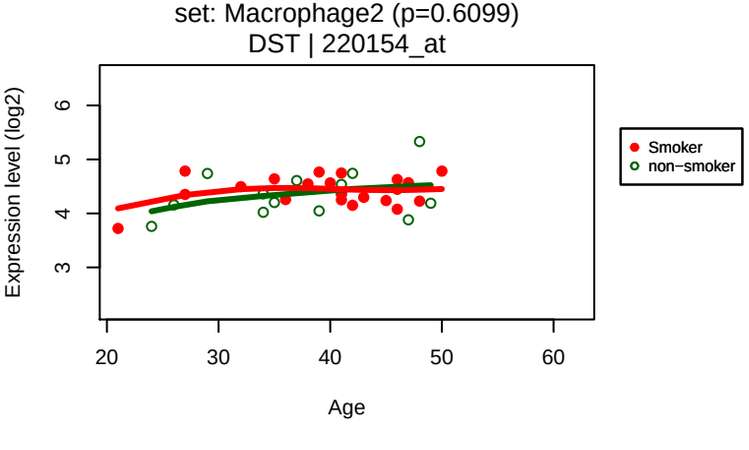

set: SmallAirway2 (p=0.0055)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker

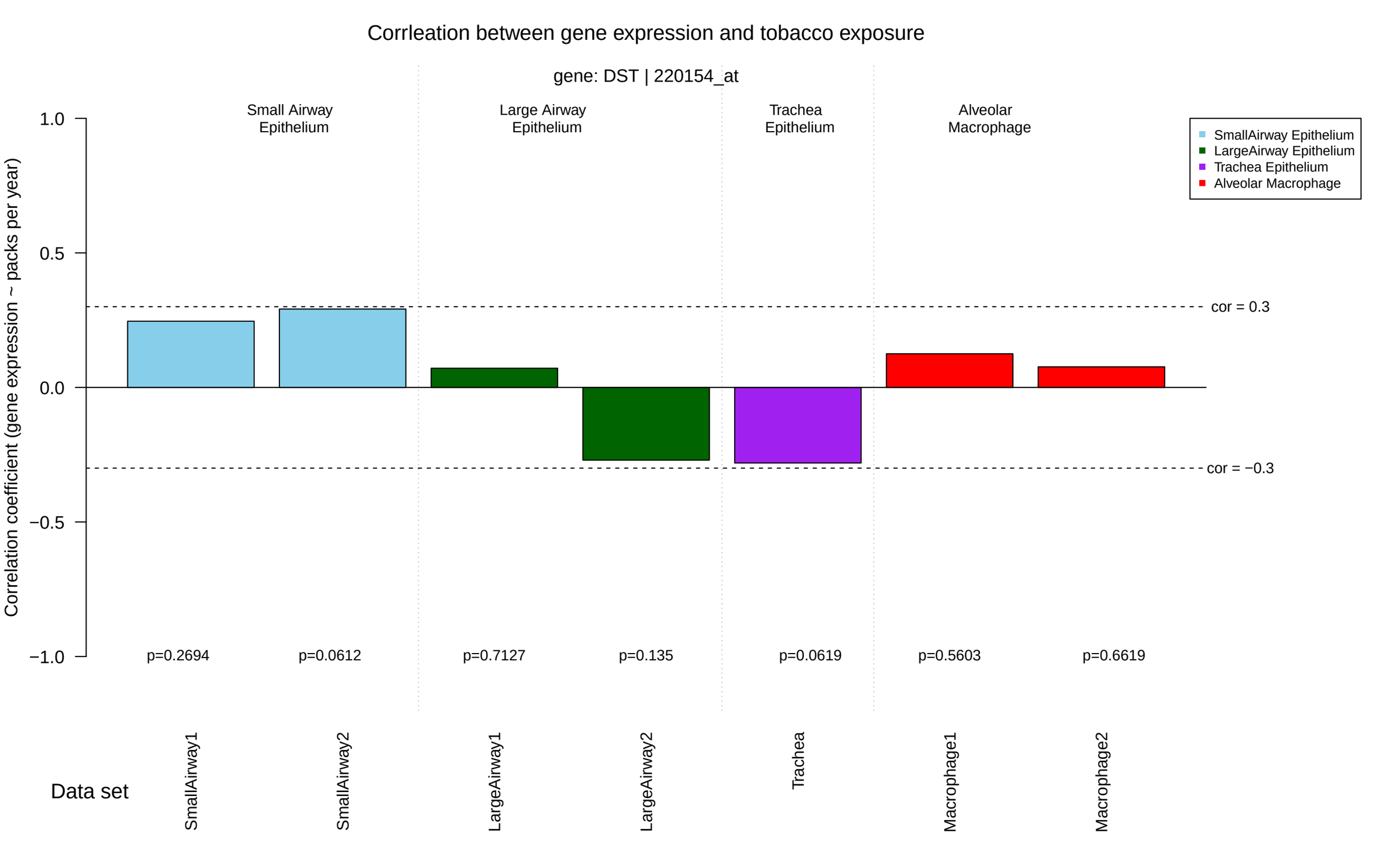

set: LargeAirway1 (p=0.7285)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2075)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0324)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2787)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6099)
DST | 220154_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DST | 220154_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2694
p=0.0612
p=0.7127
p=0.135
p=0.0619
p=0.5603
p=0.6619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2